

CN Unit 5 Question Bank — Application Layer & Security

Computer Networks (CN) · Unit 5 · Application Layer & Security · 2025

Important and frequently asked questions, grouped by marks.

Part A — Short answer (2 marks)

1. What is meant by DNS? Discuss its advantages and limitations.
2. Explain HTTP with a neat diagram and a suitable example.
3. Discuss the role of Email Protocols in Application Layer & Security.
4. What is meant by Symmetric and Public Key Cryptography? Discuss its advantages and limitations.
5. Illustrate Digital Signatures with a worked example.
6. Discuss the role of Firewalls in Application Layer & Security.

Part B — Long answer (10 marks)

1. Explain DNS with a neat diagram and a suitable example.
2. What is meant by HTTP? Discuss its advantages and limitations.
3. What is meant by Email Protocols? Discuss its advantages and limitations.
4. Illustrate Symmetric and Public Key Cryptography with a worked example.
5. Compare Digital Signatures with HTTP. When would you prefer one over the other?
6. What is meant by Firewalls? Discuss its advantages and limitations.
7. Define DNS. Derive or justify the key properties involved.
8. Define HTTP. Derive or justify the key properties involved.
9. Discuss the role of Email Protocols in Application Layer & Security.
10. Discuss the role of Symmetric and Public Key Cryptography in Application Layer & Security.
11. Write short notes on Digital Signatures.
12. Compare Firewalls with Email Protocols. When would you prefer one over the other?